

PPL LAB-7 EXERCISES

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TITLE : PROGRAMS ON STRINGS IN CUDA

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Question 1:

Write a program in CUDA to count the number of times a given word is repeated in a sentence. (Use atomic functions)

Code:

```
#include "cuda_runtime.h"
#include "device_launch_parameters.h"
#include <stdio.h>
#include <string.h>

__device__ bool checkWord(char* str, int idx, char* key, int keylen) {
    for (int i = 0; i < keylen; i++) {
        if (str[idx + i] != key[i]) {
            return false;
        }
    }
    return true;
}
```

```

__global__ void wordMatchOnString(char* str, int len, char* key, int keylen, int *count) {
    int id = blockIdx.x * blockDim.x + threadIdx.x;
    if (id < len - keylen + 1) {
        if (checkWord(str, id, key, keylen)) {
            atomicAdd(count, 1);
        }
    }
}

int main() {
    char str[100];
    printf("Enter sentence: ");
    scanf("%[^\n]%*c", str);
    int len = strlen(str);

    char key[20];
    printf("Enter key: ");
    scanf("%[^\n]%*c", key);
    int keylen = strlen(key);

    int count = 0;
    char *d_str, *d_key;
    int *d_count;

    cudaMalloc((void**)&d_str, len * sizeof(char));
    cudaMalloc((void**)&d_key, keylen * sizeof(char));
    cudaMalloc((void**)&d_count, sizeof(int));

    cudaMemcpy(d_str, str, len * sizeof(char), cudaMemcpyHostToDevice);
    cudaMemcpy(d_key, key, keylen * sizeof(char), cudaMemcpyHostToDevice);
    cudaMemcpy(d_count, &count, sizeof(int), cudaMemcpyHostToDevice);
}

```

```

    wordMatchOnString<<<ceil((float)len/256), 256>>>(d_str, l
en, d_key, keylen, d_count);
    cudaDeviceSynchronize();
    cudaMemcpy(&count, d_count, sizeof(int), cudaMemcpyDevice
ToHost);
    printf("Occurrences of the word '%s': %d\n", key, count);

    cudaFree(d_str);
    cudaFree(d_key);
    cudaFree(d_count);
    return 0;
}

```

Output:

```

STUDENT@MIT-ICT-LAB5-4:~/Desktop/230905010/07_LAB$ nvcc l7q1.cu -o l7q1
STUDENT@MIT-ICT-LAB5-4:~/Desktop/230905010/07_LAB$ ./l7q1
Enter sentence: hehe nice hehe meeting hehe you hehe
Enter key: hehe
Occurrences of the word 'hehe': 4

```

Question 2:

Write a CUDA program that reads a string S and produces the string RS as follows:

- Input string S: PCAP
- Output string RS: PCAPPCAPCP

Code:

```

#include "cuda_runtime.h"
#include "device_launch_parameters.h"
#include <stdio.h>
#include <string.h>

__global__ void buildRS(char* s, char* rs, int len) {
    int i = threadIdx.x;

```

```

    // Calculate starting offset in the output string
    int offset = 0;
    for (int j = 0; j < i; j++) {
        offset += len - j;
    }
    // Copy first (len - i) characters of s to rs at offset
    for (int j = 0; j < len - i; j++) {
        rs[offset + j] = s[j];
    }
}

int main() {
    char s[100];
    printf("Enter string S: ");
    scanf("%s", s);
    int len = strlen(s);

    int rs_len = len * (len + 1) / 2; // total output size
    char* rs = (char*)malloc((rs_len + 1) * sizeof(char));
    // +1 for '\0'
    rs[rs_len] = '\0';

    char *d_s, *d_rs;
    cudaMalloc((void**)&d_s, len * sizeof(char));
    cudaMalloc((void**)&d_rs, rs_len * sizeof(char));

    cudaMemcpy(d_s, s, len * sizeof(char), cudaMemcpyHostToDevice);

    buildRS<<<1, len>>>(d_s, d_rs, len);
    cudaDeviceSynchronize();

    cudaMemcpy(rs, d_rs, rs_len * sizeof(char), cudaMemcpyDeviceToHost);

    printf("Output string RS: %s\n", rs);
}

```

```
    cudaFree(d_s);  
    cudaFree(d_rs);  
    free(rs);  
  
    return 0;  
}
```

Output:

```
STUDENT@MIT-ICT-LABS-4:~/Desktop/230905010/07_LAB$ nvcc l7q2.cu -o l7q2  
STUDENT@MIT-ICT-LABS-4:~/Desktop/230905010/07_LAB$ ./l7q2  
Enter string S: PCAP  
Output string RS: PCAPPCAPCP
```